

The Self-Consistent D -Composite Scheme at Linear Order: Sector Constraint, Symbol Identity, Off-Resonance Flatness, and the Six-Dimensional Resonant Sector

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Abstract

We begin the audit of the last unanalyzed variational scheme of the minimal branch: the self-consistent D -composite system in which the connection sits inside the defining jet. Four exact results are proved at the linearized, frozen-coefficient level around the affine point. (i) Imposing $\Theta \in W_L$ defines a consistent Lorentz-real subsector; necessity for self-consistent solutions is *not* proved (a constant Hermitian shift at a constant tetrad is a counterexample). (ii) The frozen symbol $A(s, \lambda)$ of the linearized implicit system satisfies the exact operator identity $A^3 = (\lambda \cdot s) A^2$; its spectrum is $\{0^{10}, q^6\}$ with $q = \lambda \cdot s$, with three nilpotent Jordan blocks on the kernel and $\det(I - A) = (1 - q)^6$. (iii) Off the single resonance hyperplane $q = 1$ the unique $\delta\Theta$ -driven solution is the exact gradient; the linearized tetrad is holonomic and the linearized geometry pullback-flat. (iv) At $q = 1$ there is an exactly six-dimensional space of homogeneous resonant modes — $\dim \mathfrak{so}(1, 3)$ of them — and every one is anholonomic with linearly independent curls. All linearized anholonomy, hence all candidate linearized curvature, is confined to this resonant sector. The variable-coefficient assembly of resonant modes, the quadratic action on the resonant sector, on-shell torsion, and the chain-rule spin current remain open. All algebraic statements are verified in `tools/verify_dcomposite_linearized.py`.

1 The self-consistent system

The scheme couples the covariant jet back to the connection:

$$E_\mu = \mathcal{N}_0^{-1/2} (\partial_\mu \Theta + \Omega_\mu[E] \Theta + \Theta \Omega_\mu[E]^\dagger), \quad g_{\mu\nu} = \frac{1}{2} (E_\mu^\# E_\nu + E_\nu^\# E_\mu), \quad (1)$$

with $\Omega_\mu[E]$ the spin representation of the Levi–Civita connection of the tetrad defined by the slice coordinates of E_μ (torsion-free branch first; anholonomy, not torsion, is the requirement for curvature).

Lemma 1 (W_L subsector). *Demanding $E_\mu \in W_L$ for every independently chosen Lorentz connection forces the Hermitian part of Θ to vanish; hence $\Theta \in W_L$ defines a consistent Lorentz-real subsector (exact solve; check L0). This does not prove necessity for self-consistent solutions, where only $\Omega = \Omega[E]$ occurs: $\Theta = \Theta_{\text{aff}} + H$ with constant Hermitian H has $\Omega[E] = 0$ and $D_\mu \Theta \in W_L$ although $\Theta \notin W_L$ (check L0b). All results below are statements about the W_L subsector.*

2 Linearization and the frozen symbol

The affine field $\Theta_{\text{aff}} = \Theta_0 + \sqrt{\mathcal{N}_0} \bar{E}_\nu x^\nu$ with $\Theta_0 \in W_L$ is an exact solution of (1) with $\Omega = 0$. Linearize $\Theta = \Theta_{\text{aff}} + \varepsilon \theta$, $E = \bar{E} + \varepsilon F$, freeze coefficients at x_0 , and pass to the real-exponential (Laplace-type) symbol $\partial_\nu \rightarrow s_\nu$ with real s ; for real Fourier modes $\partial_\nu \rightarrow ik_\nu$ the invariant is

$q = i\lambda \cdot k$, so $q = 1$ has no real- k solution and the sector below consists of exponential/evanescent symbol modes. The system becomes

$$(I - A(s, \lambda)) F = \partial\theta', \quad \lambda^a = \text{slice coords of } \Theta_{\text{aff}}(x_0)/\sqrt{\mathcal{N}_0} = x_0^a + \theta_0^a, \quad (2)$$

where A implements the linearized Levi-Civita connection of the perturbed tetrad acting on the background through the bimodule. $\mathcal{N}_0$ cancels; A is bilinear in (s, λ) . Under the diagonal Lorentz group at the identity-tetrad background the mixed contraction $q = \lambda^a s_a$ is invariant.

Theorem 1 (Symbol identity and spectrum). *$A^3 = q A^2$ holds as a polynomial identity in all eight variables (check D2, symbolic). Generically $\text{rank } A = 9$, $\text{rank } A^2 = 6$; hence the characteristic polynomial is $t^{10}(t - q)^6$, the minimal polynomial $t^2(t - q)$, and $\det(I - A) = (1 - q)^6$ (check D3). The frozen full symbol is modewise invertible iff $q \neq 1$; local or global uniqueness for the variable-coefficient PDE does not follow from this alone.*

Theorem 2 (Off-resonance flatness). *The linearized Levi-Civita connection factors through the curl of the tetrad perturbation, so A annihilates every exact gradient (check D1). Consequently, for $q \neq 1$ the unique $\delta\Theta$ -driven solution of $(I - A)F = \partial\theta'$ is $F = \partial\theta'$ itself: exactly holonomic, with pullback-flat linearized geometry (check D4).*

Theorem 3 (Resonant sector). *At $q = 1$ the eigenspace $\ker(I - A)$ is exactly six-dimensional ($= \dim \mathfrak{so}(1, 3)$). Every resonant mode is anholonomic and the six curls are linearly independent (check D5, exact at a generic resonant point). All linearized anholonomy of the D -composite scheme is confined to this sector.*

Corollary 1 (Where linearized curvature can live). *Since curvature at linear order requires anholonomy, any linearized gravitational content of the D -composite scheme must be assembled from the exponential symbol modes on the moving surface $(x_0 + \theta_0) \cdot s = 1$; each such mode carries a nonzero linearized Riemann image (check D5c), but their relation to real-frequency propagation is open. Off it, the scheme is flat at linear order — but, unlike the exact-gradient restriction, not identically: the resonant sector is a genuine nontrivial remainder.*

3 Answers to the six audit questions at this level

- (1) Modewise invertibility of the frozen full symbol: yes iff $q \neq 1$; PDE-level uniqueness open.
- (2) Anholonomy: driven solutions are exactly holonomic; anholonomy exists only in the resonant sector.
- (3) $R \neq 0$ without inserting curvature by hand: at linear order only through resonant modes; whether they assemble into curved solutions of the variable-coefficient system is open.
- (4) Chain-rule spin current: not yet computed.
- (5) On-shell torsion selection: not yet computed.
- (6) Degeneracy: off resonance the scheme degenerates to the flat sector, but not identically — the six-dimensional resonant remainder is the irreducible core on which the quadratic action must be evaluated next.

Scope and non-claims

Frozen-coefficient (microlocal) analysis at the affine point. Not a solution theory for the variable-coefficient system: λ moves with x_0 , so the resonance set sweeps spacetime and the constant-coefficient solvability statements do not globalize automatically. Nothing here derives the Hilbert-Palatini term, decides on-shell torsion, or establishes curved solutions. A discovered-and-fixed index transposition in the linearized connection (which leaked the symmetric tetrad part into ω^1) is documented in the verifier as regression check D1; results computed with the faulty formula (an earlier $\{0^4, q^9, (q/2)^3\}$ spectrum) are void.